Rising hot nights, persistent cold days, and an aging population: thermal extremes and cardiovascular hospital burden in Hong Kong, 2013–2023

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Laidlaw Scholars · Ecological monthly time-series ($N = 132$) · REAL HKO & C&SD · HA outcomes pending HPC · No synthetic coefficients as findings · ISO A0 (1189 mm × 841 mm)

1. Why climate extremes matter for human health

Climate change reshapes thermal exposure in subtropical cities. Overnight heat impairs physiological recovery; cold snaps stress blood pressure, viscosity, and autonomic tone—raising acute cardiac and cerebrovascular risk among older adults in dense housing with limited overnight cooling.

- Official HKO metrics drive public heat/cold warnings and preparedness
- Ages 65–74 nearly doubled (2013–2023), multiplying vulnerable person-time
- Hospital burden—not only mortality—matters for adaptation planning

2. Literature synthesis (Hong Kong & related)

Cold-dominant CVD (daily).
- **Goggins 2013** (AMI 2000–09): below ∼24°C, **∼+3.7% AMI / °C** cooling (lags 0–13); **heat NS**; NO_2 key.
- **Goggins 2012** (stroke, age 35+): HS **∼+2.7%/°C** (lags 0–4); IS **∼+1.6%/°C** below ∼22°C (lags 0–13).

Night heat: intensity ≠ binary flag.
- **Guo 2024**: official HN (Tmin ≥ 28°C) **null** for excess hosp.; hot-night **intensity ∼+3.1%** (circulatory sensitive).
- **Wang 2019**: ≥5 consecutive HNs and 2D3N combinations raise mortality.

Modifiers & warm-climate winter excess.
- **Guo 2025**: temp.–hosp. risks amplified on polluted days.
- **Yang/Chong 2025**: cold + influenza ↑ stroke (∼1.17M; 22y).
- **Lam 2018**: stronger cold–AMI in diabetes; **Lorking 2020**: tropical winter stroke excess.

3. Mini evidence table (estimands not interchangeable)

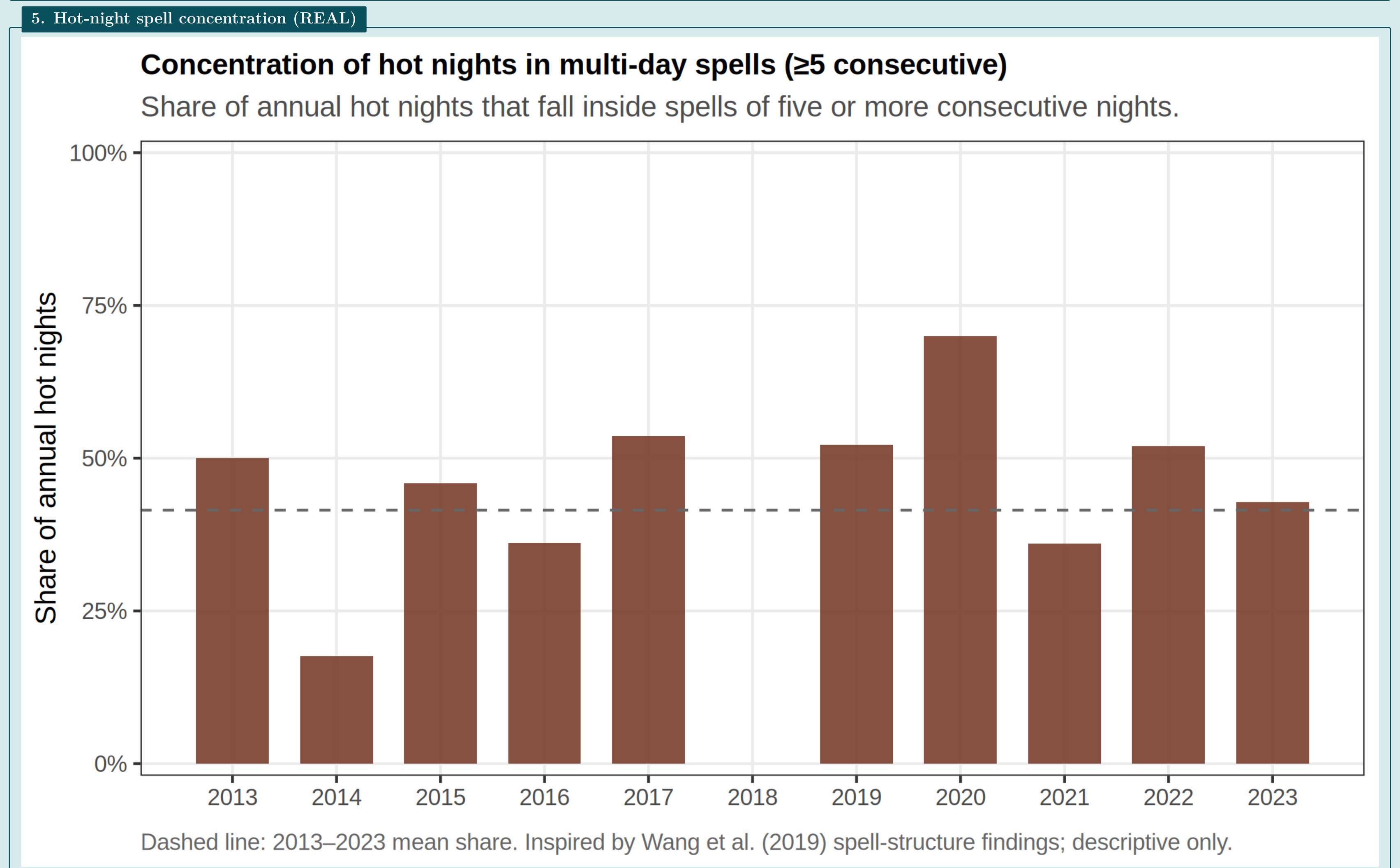
Study	Key finding	Scale
Goggins 2013	AMI +3.7%/°C (<24°C); heat NS	Daily
Goggins 2012	HS +2.7%; IS +1.6%/°C	Daily
Guo 2024	Binary HN null; HNe ∼+3.1%	Daily
Wang 2019	Spell / 2D3N informative	Daily
Yang 2025	Cold + ILI ↑ stroke	Weekly
Lam 2018	Cold–AMI stronger in DM	Daily
This project	RR per 5 extreme days (planned)	**Monthly**

Prior HK cold–CVD band ≈ 1.6–3.7% per °C (daily). Do **not** compare to planned monthly RR per 5 extremes.

4. Aim, design & contribution

Aim. Monthly associations of official hot nights & cold days with principal-dx AMI, ischemic and hemorrhagic stroke (age 35+; 2013–2023) with age–sex offsets.
Design. Ecological time-series, $N = 132$ months; population-level only.
Contribution. Monthly policy-metric audit for diagnosis-specific AMI/IS/HS with aging, COVID handling, staged pollution—**coexistence** framing. Not "first hot-night hosp. study" (Guo 2024).

5. Hot-night spell concentration (REAL)



Share of hot nights in multi-night spells rose from ∼34% (2013–18) to ∼51% (2019–23).

	Goggins 2012/13	This project
Resolution	Daily DL/GAM	Monthly counts
Exposure	Mean °C	Official extreme days
Effect	% per 1°C	RR per 5 extremes
Focus	Cold-dominant CVD	Heat + cold coexistence

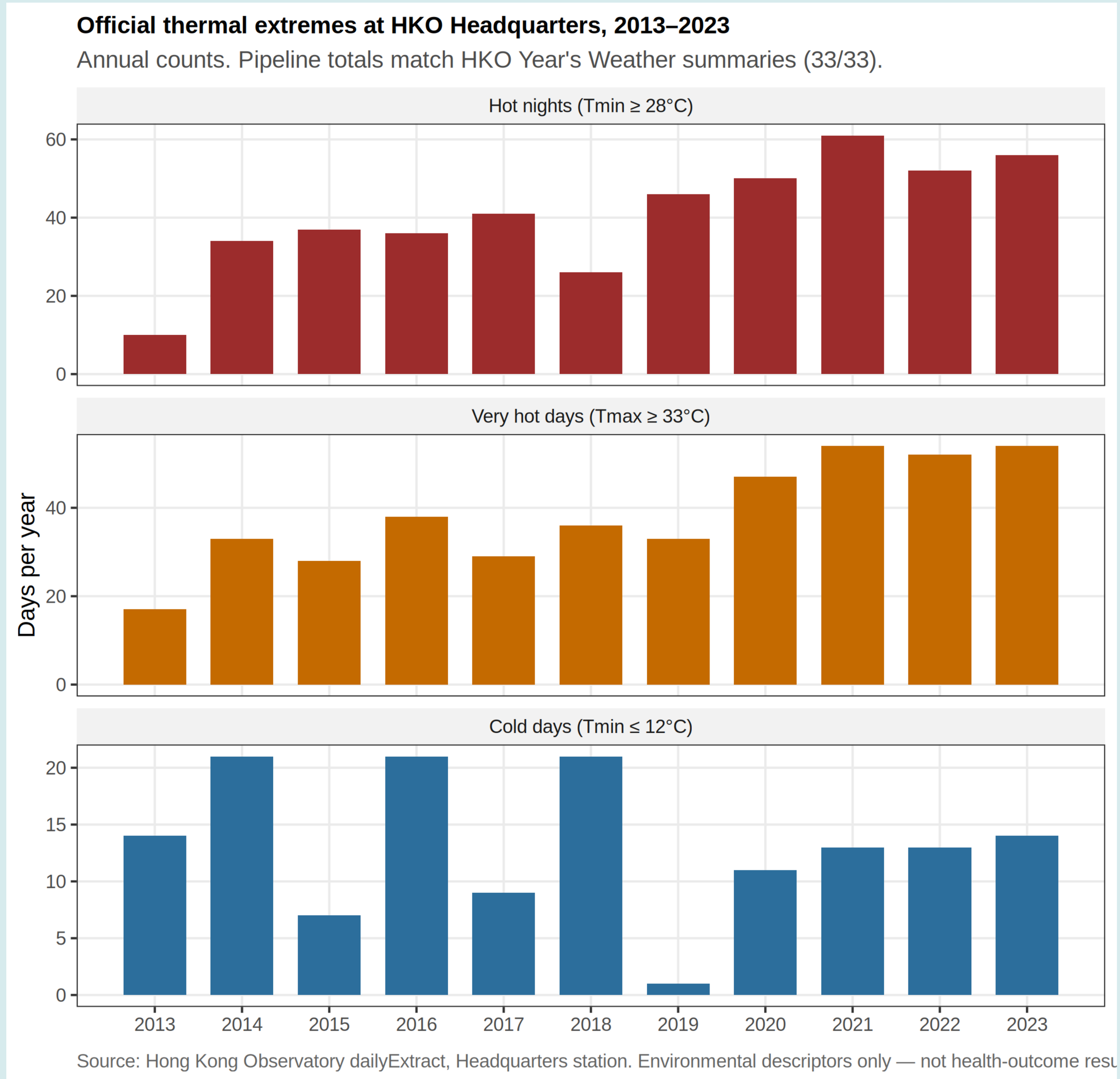
Daily = acute **triggering**; monthly = hospital **burden**. Coefficients are not interchangeable.

6. Data sources & official exposures

Domain	Source / status
Weather	HKO Headquarters daily **(REAL; 33/33 validated)**
Population	C&SD 110-01001 mid-year age×sex **(REAL)**
Pollution	EPD NO_2/O_3/PM (import ready; staged)
Outcomes	HA CVD admissions **(pending HPC)**

Hot night Tmin ≥ 28°C; VHD Tmax ≥ 33°C; extreme hot Tmax ≥ 35°C; cold Tmin ≤ 12°C. Primary: monthly HN; co-primary: cold days; spell-ge5 & 2D3N alternatives.

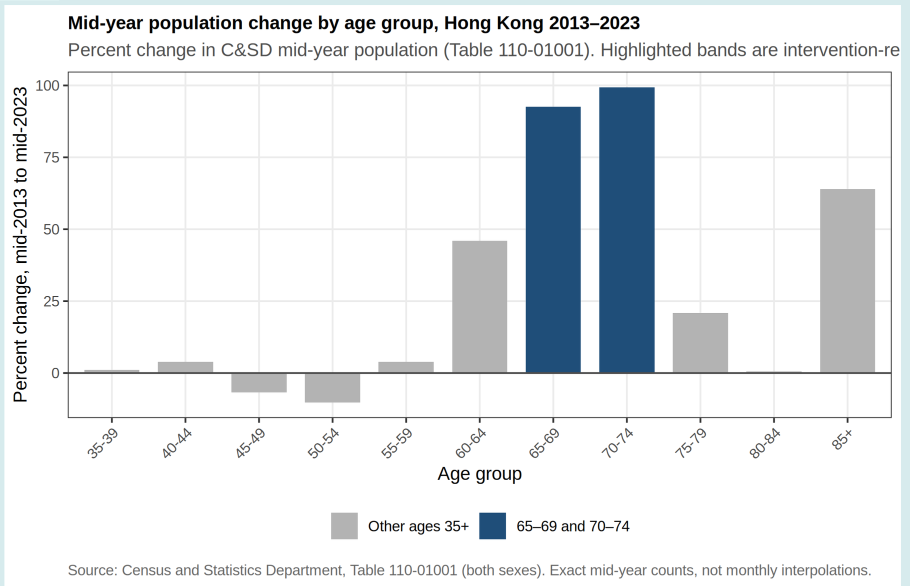
7. Rising heat, persistent cold (REAL)



Environmental descriptors only. Pipeline totals 33/33 MATCH vs HKO *Year's Weather*.

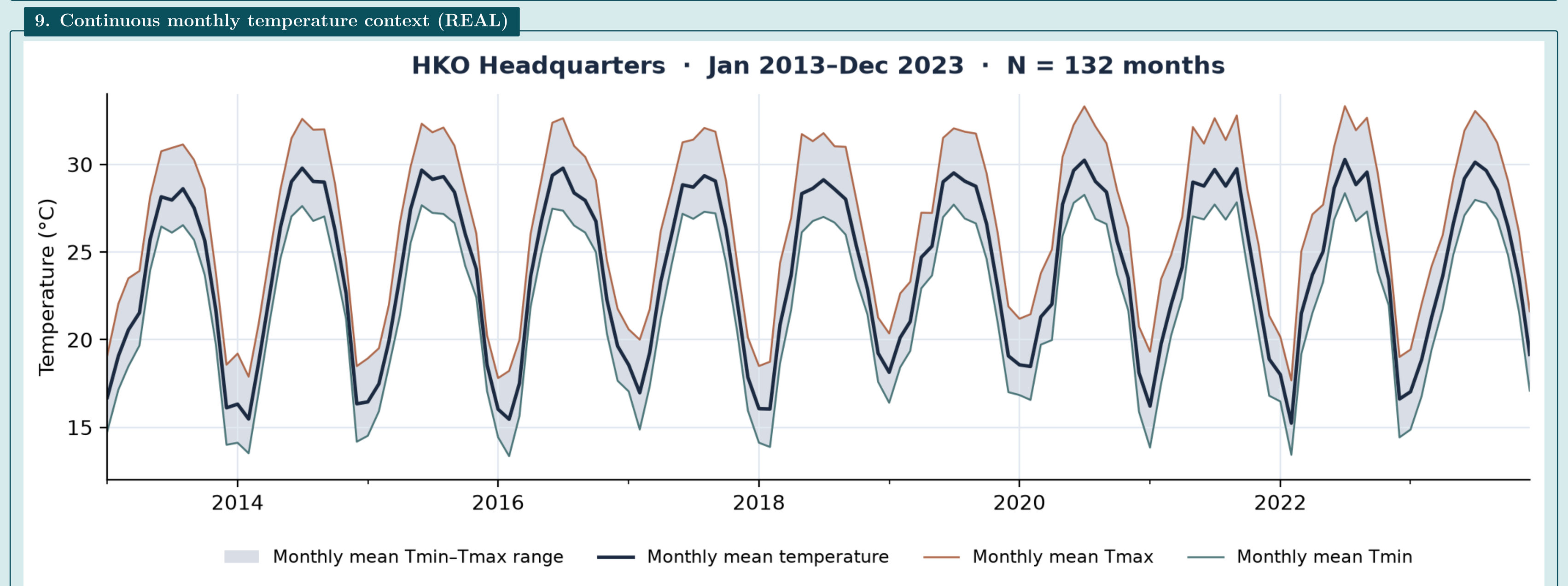
- HN: **10 (2013)→61 (2021)/56 (2023)**; mean 30.7→53.0/yr
- VHD mean 30.2→48.0/yr; cold days persist 13–14/yr (2021–23)
- HN in ≥5-night spells: 34%→51%; 2D3N months/yr: 2.7→4.4

8. Aging multiplies human stakes (REAL)



- **65–69 +92.6%** | **70–74 +99.3%** | 85+ +63.9% (C&SD mid-year)
- Separating 65–69 and 70–74 is clinically and operationally essential for HA planning

9. Continuous monthly temperature context (REAL)



HKO Headquarters monthly mean with Tmin–Tmax ribbon; $N = 132$ months. Exposure file ready for HA merge.

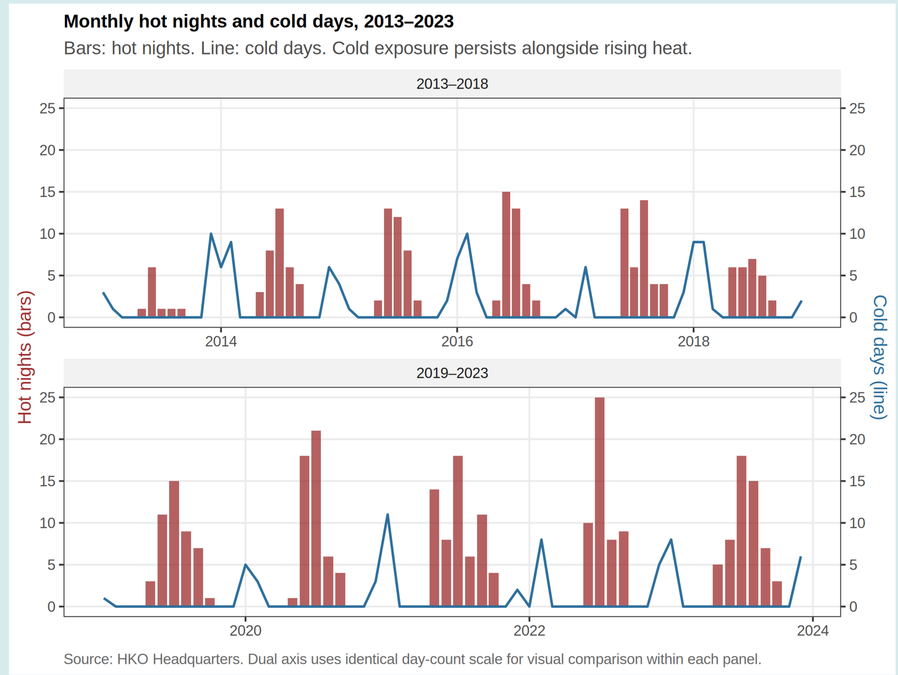
10. Seventeen analysis pathways

Dev/real tracks: synthetic HA dry-runs only vs refused synthetic outcomes.

#	Pathway	Status
00	Project setup & session info	Done
01	Import HKO daily weather	Done
02	Build monthly climate extremes	Done
03	Import / document EPD pollution	Ready
04	Build monthly pollution panel	Ready
05	Build C&SD age–sex denominators	Done
06	Confounders (COVID, holidays)	Done
07	Synthetic HA dry-run outcomes	Dev only
08	Merge analysis panel (dev)	Done
08b	Merge approved HA panel (real)	Pending
09	Descriptive tables	Done
10	Exploratory figures	Done
11	Fit NB rate models (syn./real)	Scaffold
12	Model diagnostics (syn./real)	Scaffold
13	Report outputs & status note	Done
14	Validated annual extremes figure	Done
15	Smoke checks (HKO 33/33)	Done
16	Exposure–aging deliverable	Done
17	Lab-meeting temperature figures	Done

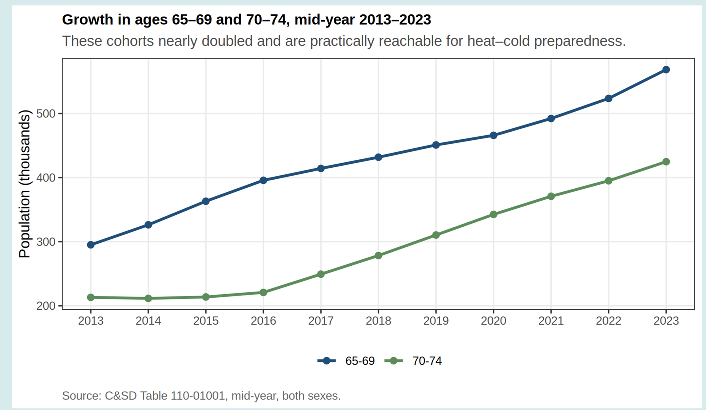
run_pipeline_dev.R / run_pipeline_real.R. Labels: REAL / PLACEHOLDER / SYNTHETIC.

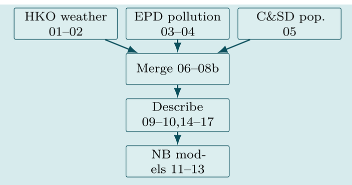
11. Monthly hot nights vs cold days (REAL)



Bars = hot nights; line = cold days. Cold exposure persists alongside rising heat.

12. Older cohorts & pipeline logic





13. Planned analysis & caveats

When HA is live (ICD-9; patient-level; ages 65–69 & 70–74 separable):
- Month × age × sex × diagnosis aggregates
- NB rates; log(population × days) offset
- HN & cold days (per 5); Tmax/Tmin + lag; COVID phases; staged pollution; cautious O_3
- Outcomes: AMI, IS, HS (principal diagnosis)

No synthetic coefficients as findings.

Caveats. Null official HN compatible with Guo 2024; monthly = burden not triggering; HQ T is a territory proxy; no cold→heat regime-shift claim without harmonized methods.

14. Key numbers & take-homes

10 → 61	Hot nights/year (2013 → 2021)
30.7 → 53.0	Mean HN/year (2013–18 vs 2019–23)
+92.6% / +99.3%	Pop. ages 65–69 / 70–74
33/33	HKO Year's Weather matches
132 months	Jan 2013–Dec 2023

1. Heat rose after 2018; **cold days did not disappear**.
2. Near-doubling of ages 65–74 multiplies CVD person-time at risk.
3. A reproducible 17-step platform is ready for HA hospital-burden estimation.

Next: HPC → HA QC & Table 1 → pre-specified NB models.
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